Moumen Nermine

Final-year Industrial Chemistry Engineering Student

• *nermine.moumen@insat.ucar.tn* • +21627040414 • *linkedin.com/in/nermine-moumen*

EDUCATION

National Institute of Applied Sciences and Technology (INSAT) 2022–Present

Industrial Chemistry student, specializing in Process and Environmental Engineering, with a solid training in chemical catalysis, electrochemical kinetics and materials science, complemented by foundations in analytical chemistry, organic chemistry and laboratory HSE. Interested in green chemistry, biomass valorization and catalytic materials. Seeking a 6-month final-year internship.

PROFESSIONAL EXPERIENCE

SEGULA Technologies - Internship in Electrochemical Modeling of Batteries 06/2026–08/2026

- Modeling of a **Li-ion** cell (NMC811/Graphite-SiOx) using the Single Particle Model (SPM) implemented in MATLAB.
- Optimization of the mesh and diffusion coefficient values, reducing the **RMSE from 202.13 mV to 40.5 mV**.
- Validation of the results through an iterative approach and a critical analysis against experimental data and COMSOL.

Control Union - Quality Control Internship 08/2025–09/2025

- Analysis of lubricants (viscosity and heavy metals by **ICP-OES**).
- Preparation and interpretation of analysis reports to verify compliance with **ASTM D5185** and **ISO 17025** standards.

ETAP - Internship in Analytical Characterization of Hydrocarbons and Water Treatment 07/2024–08/2024

- Training in organic geochemistry techniques (GC, GC-MS, Rock-Eval pyrolysis) for the characterization of hydrocarbons.
- Physico-chemical analyses of water (pH, salinity, bicarbonates, sulfates, chlorides) by ion chromatography and ICP-AES.

ACADEMIC PROJECTS

Molecularly Imprinted Polymers (MIPs) for Pharmaceutical Applications

- Synthesis and characterization of a sulpiride-selective MIP by radical polymerization, with Soxhlet extraction, monitored by UV-Vis spectrophotometry and validated by controlled-release kinetics (240 min).
- Synthesis of a MOF/polymer core-shell MIP for targeted delivery of doxorubicin, with an **85% encapsulation efficiency**.
- Modeling of **adsorption isotherms** (bi-Langmuir, Freundlich) to relate material properties, binding sites and performance.

Techno-Environmental Analysis of Ethylene Production: Steam Cracking vs Bioethanol Dehydration

- Simulation of both routes in **Aspen Plus V14**, including bioethanol dehydration over an **HZSM-5 acid catalyst**.
- Cradle-to-gate **LCA** (OpenLCA, ISO 14040/14044) on 5 impact categories; identification of the origin of the biomass and indirect land-use change as the main uncertainties.

Literature Review: Synthesis and Characterization of Superconducting Materials

- Analysis of the fundamental properties of superconductors and study of fabrication routes: solid-state ceramic synthesis, crystal growth (flux, Bridgman, Czochralski), and thin-film deposition (ALD).
- Study of characterization methods: RAMAN, FTIR, XRD, SEM and TEM.

Prediction of Liquid–Liquid Equilibria and Cross-flow and Countercurrent Extraction for Ternary Systems

- Formulation of the LLE thermodynamic system and resolution via Rachford-Rice and arc-length continuation.
- Implementation of the workflow in Python, MATLAB and Mathematica (spline/fsolve, SciPy/mpltern, Interpolation/FindRoot).

TECHNICAL SKILLS

- **Lab Techniques:** Chromatography (HPLC, GC, TLC), Spectrometry (MS, ICP-OES), Spectroscopy (NMR, FTIR, UV-Vis), Diffraction (XRD), Microscopy (SEM, TEM), Electrochemistry (Cyclic voltammetry, Polarography).
- **Simulation:** Aspen Plus, Aspen HYSYS, COMSOL Multiphysics • **Programming:** *MATLAB, Mathematica, Python*
- **Data & Office:** Microsoft Office Suite, OpenLCA • **Standards:** ISO 9001, ISO 14001, ISO 45001, ISO 17025, ISO 14040

EXTRACURRICULAR ACTIVITIES

IEEE INSAT — Organizing Chair of TRSYP 3.0; Social Activities Chair – MENASYP; Chair of EMBS.
THEATRO INSAT — Member of the dance team.

LANGUAGES AND CERTIFICATES

Languages
English (Advanced), French (Bilingual), Arabic (Native).

Certificates
Aspen HYSYS – MATLAB